

User Perceptions of Defensive Techniques Against Keystroke Timing Attacks During Password Entry

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ABSTRACT

To protect the password from visual attacks, most password entry screens use a password masking scheme that displays a series of placeholder characters (e.g., dots and asterisks) instead of the actual password. Recent research has however shown the security provided by this form of password masking to be weak against keystroke timing-analytics attacks. The underlying idea behind these attacks is that, even when a password is masked as described above, the timing between consecutive placeholder characters gives away information about the password since the relative locations of characters on the keyboard dictate how fast fingers move between them. In this paper we argue that, for security-sensitive applications, password masking mechanisms ought to hide the true intervals between password characters in order to overcome these kinds of attacks. Making adjustments to these timings however has the potential to pose usability issues given the fact that the typing would not perfectly align with the display of typed content. The paper proposes 3 different password masking schemes and undertakes a usability evaluation on them. Our early results suggest that user receptiveness to two of the schemes is not much worse than that seen with the conventional (insecure) scheme.

CCS CONCEPTS

• **Security and privacy** → *Browser security*; **Biometrics**; **Authentication**.

KEYWORDS

Password security, password usability, keystroke timing attacks, shoulder surfing

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1 INTRODUCTION

During password entry on computing devices, most input screens replace a user's actual password with a sequence of placeholder characters (e.g., dots and asterisks)¹. These placeholder characters are meant to serve two purposes: (1) privacy — to protect the password from visual attacks, and, (2) usability — to enable the user to determine if a given key has been registered, which in turn prevents the user from making mistakes.

Given an adversary who is able to precisely measure the timing between consecutive placeholder characters during password entry, recent research has shown the privacy argument to be subject to question. For example, in scenarios where the password entry process is projected onto secondary displays (e.g., projector screens or conference call videos such as via zoom), an adversary could easily record a video of the screen and then later analyze the time-intervals between placeholder characters [1]. Leveraging well-known patterns of the time-intervals associated with different digraphs [11], such an adversary could then make clever guesses on which characters might be part of the password. For example, a time-interval of about 90 milliseconds between two keystrokes has a higher probability to represent the digraph “vo” than the digraph “vb” [12] (See [11] for timing statistics of more digraphs). Recent research (e.g., Balagani *et al.* [1]) has shown this line of attack to significantly improve an attacker's odds of determining the password.

In light of this privacy threat, this paper makes the case for a new approach to the password masking process that is designed to mitigate this family of timing attacks. We propose that the time-intervals between placeholder characters used for password masking should not perfectly align with those of the true password's characters. A modification of this kind to the password entry process would of course come with a whole range of new usability questions. For example, if an asterisk mapping to a given character is delayed too much, the user could assume the character was not registered by the system. Such a user could retype the character,

¹We alternately use the term *password masking* to refer to this process.

only to later realize that the placeholder was just delayed. On the other hand if a defense mechanism simply eliminated the display of placeholders (or asterisks), the user might just not be sure of the password was entered at all. The paper examines this line of attack on password entry with emphasis on the usability of password masking schemes designed to thwart the attack. The paper makes *two primary contributions*:

Password masking schemes: We present three password masking schemes that distort the digraph-timing correlations exploited by the above described attack. In two of the schemes, we perturb the timing dynamics of the placeholder characters using two different strategies, a user-tailored one, and a user-independent one. In a third scheme, we eliminate the placeholder feedback completely (i.e., similar to the scheme used on linux terminals). To provide a basis to evaluate the impact of the mitigations to the attack, we replicate the attack and evaluate its performance under the defense mechanisms.

User receptiveness to password masking schemes: We conduct a usability experiment to study users' attitudes towards the three schemes as compared to the conventional approach being used on most of today's system. While one of the schemes (the linux-centered one) is already used in the linux terminals, no prior work has to our knowledge studied its usability in GUI settings for which the end-user might not be an expert (linux terminals tend to be used by specific types of technical experts and hobbyists). Further, our study examines it side-by-side with the other options, providing information on how it would fare as an option of choice against the attack in standard password entry screens. We find two of the schemes to be well-received by the users who rated them favorably with the current (vulnerable) standard.

2 RELATED WORK

The privacy threat posed by timing analytics on the placeholders displayed during password entry was first studied by Balagani *et al.* [1]. The authors in that paper coined the acronym, PILOT to refer to the attack. We will hence refer to it as the PILOT attack in the rest of the paper. To showcase the PILOT attack, the authors conducted two different experiments: one based on passwords entered on a laptop, the second based on PINs entered on a terminal designed to closely mimic an Automatic Teller Machine (ATM). In the password-based experiment, the laptop on which participants entered several passwords was connected to a projector whose display was recorded by a camera. In the PIN-based attack, a video camera directly recorded the screen of the ATM-like terminal. In both cases the pattern of asterisks as captured by the video camera were then used — in conjunction with a machine learning model trained using pre-existing keystroke dynamics datasets — to drive the attack to infer the PINs/passwords. Both attacks were shown to be highly successful. The laptop attack reduced the number of attempts to guess a password by between 25% and 385%, depending on the password. While the PIN attack used the timing information to guess around 7% of PINs within 40 attempts, compared to 0.4% without. The commonality between our work and Balagani *et al.* is that we study the same line of attack. However, while Balagani *et al.* were focused on showcasing the attack, our work is focused on

studying changes to the password entry process that protect the password from the attack.

A significant amount of past research has put forth schemes that seek to ensure privacy of the password while it is entered into a computing device. The majority of these works focus on attacks that leverage direct observation of the characters constituting the password. The adversary executes such attacks by either observing the fingers during the typing process, or the buttons of a virtual keyboard as they get highlighted during typing. For example, De Luca *et al.* [3] designed a scheme which displays fake cursors of various colors on a virtual keyboard during the password typing process. When the user taps a key such as "A", a cursor (or selector arrow) is displayed at letter "A", as well as at several other letters on the keypad. An attacker seeing all these cursors on the virtual keyboard would not know which one maps to the actual keys being typed. In [2], a scheme based on vibration feedback was designed to obfuscate the password via the insertion of fake characters into the password string. When a user types a password under this scheme, a subset of characters being typed vibrate, while others do not. When a character vibrates when tapped, the user takes that as a cue that the next character to be entered should be fake. Hence ultimately the password being entered is a mix of real and fake characters. An attacker observing the password entry process would not know which characters are associated with vibrations, and hence would not know the actual password.

The works cited above share with us the overall objective of protecting the password entry process. That said, our work seeks to mitigate a completely different and more recently discovered line of attack. In particular, our work seeks to mitigate *keystroke timing analytics attacks* while the body of previous work seeks to mitigate *direct visual attacks on the characters* comprising the password. The concern of our work is on ensuring that the temporal patterns of the symbols (e.g., asterisks) displayed to the user on conventional password entry screen designs does not leak the password. The above difference notwithstanding, it could be argued that some of the above cited prior works might also present as solutions to the attack studied in this work. We note however that a key design requirement for our solution was to keep the conventional, widely adopted, text-based password entry system as intact as possible (i.e., tweak it only minimally). This further puts us apart from the above cited schemes (and other related works in the same family) given that many of them make wholesale changes to the conventional password entry mechanism (e.g., by incorporating extra hardware [10], adding new feedback modalities [2, 10] or moving towards a graphical password system [13]). Such bold changes mean that systems already in deployment would require a major overhaul, lowering the likelihood of widespread adoption in typical consumer systems.

3 PROPOSED DEFENSIVE APPROACHES AGAINST THE PASSWORD ATTACK

By default, the placeholders or asterisks are displayed immediately after a key is typed. This means that an attacker who is viewing the asterisks (e.g., on a projector screen or zoom call) can precisely determine the time-interval between every pair of keys typed and

use this information to run the attack. In Figure 1(a) this is represented by the way the real keys and the asterisks are aligned. In this paper we consider defenses against this attack based on two general themes whose description follows.

Defensive Approach #1: Removal or Scrambling of Visual Cues: Our first defensive approach is the removal of all visual cues when a user is typing their password. Removing the visual cues would make it impossible for an attacker to use video to record the timing between key presses.

Defensive Approach #2: Manipulating the Display Rate of Asterisks As opposed to entirely removing the placeholder characters, the second defense approach manipulates the time-intervals between them. Theoretically speaking, the ideal manipulation approach to completely overcome the attack is to randomize when the placeholders display. This approach would however be so disruptive to the user who might encounter a vastly different system each time they enter the password. We propose two variants of the display manipulation approach that we show to hold well under attack despite not necessarily reaching the theoretical limits offered by a random display. These variants are: (1) Defensive Approach #2a — For a pair of keystrokes separated by a time-interval less than a threshold, T_h , this approach uses a static delay, T_h , (for all users) between placeholder characters (e.g., 200 milliseconds in Figure 1(b)). For digraphs having inter-key delays less than T_h , this scheme completely eliminates any connection between the password inter-key times and the placeholder inter-key times. This defense might not always cover all characters being typed. If two characters are normally separated by a period of time longer than the target threshold, then the attacker would still be able to rank the possible key pairs for the keys being pressed. Even though the scheme only obfuscates a fraction of the keys pressed, a user would significantly mitigate the risk posed by the attack. We explore several values of this fixed interval against the attack (in Section 5) before choosing one for our experiments. (2) Defensive Approach #2b — This uses a static user-specific delay determined by the typing speed of the user (in this paper, the mean of their typing speed). The advantage of this approach relative to the previous one is it will ensure that a higher percentage of the user's inter-key times are hidden by the minimum length of time between displayed characters.

4 DATA COLLECTION PROCEDURES

Our usability dataset consists of a combination of qualitative and quantitative data to measure the usability of our baseline password input method (normally masked keys) against our safer alternatives. For this we built a website that collected background metrics to estimate usability as well as asking explicitly for user feedback. The remainder of this section will describe the website and data collection procedures in greater detail.

4.1 Website Overview

After some iterations, our website had the following sets of characteristics and features. It was a simple website with buttons for moving onto the next part of the task. The starting page of the site has details about the task if the participant needs a reminder. Second page is a brief typing task where participants type their assigned password multiple times so we can measure their typing

speed. All remaining pages are each dedicated to one of the password entry methods (in a random order) along with a 7-point Likert scale question for rating the ease of use of that password entry method. The specific question we asked participants after typing their password using each approach was, “On a scale from 1 (Very uncomfortable) to 7 (Very comfortable), how would you rate your comfort when using this method?”.

The website uses a custom password entry box since by default the HTML password entry field type does not allow the customization we needed. Although if participants make a mistake they can just try again, we track the number of attempts for each password entry type and each session as a secondary usability metric. Our four password entry methods for the usability study can be described as follows. (1) “No-Defense” — Asterisks are displayed as normal, this is the vulnerable approach to typing a password. (2) “Defense #1” — No characters are displayed on the screen as the user types. (3) “Defense #2a” — All typed characters have a minimum amount of time (250 milliseconds) that must pass between when two characters appear on the screen, this amount of time is standard for all users. Note that we picked 250 milliseconds because it covered approximately 90% of the down-down time (DDT) values in our dataset and thus is representative of a delay that would protect the majority of users. (4) “Defense #2b” — Each user has their own minimum threshold for the time between characters being displayed on the screen, this is determined by analyzing their typing speed and finding the average delay.

4.2 Data Collection Procedures

After receiving IRB, approval we recruited 18 students as volunteers (who were not compensated) for the online-only usability experiment. There were an equal number of male and female participants, with 11 majors represented among the population. Participants were asked to conduct 8 sessions of the experiment on separate days. For evaluating the usability of different password entry methods we used four different passwords. They are “123brian”, “jillie02”, “lamondre”, and “william1”. We selected these passwords because they were used for the attack in [1]. Each participant was randomly assigned a password that they used throughout all of their experimental sessions (on each of the 8 days) and during each session all of the different password entry methods have to be used and rated by the participant.

For each user, we first conducted the experiment remotely and walked them through it ourselves before having them do the experiment and ask any questions they had. We found this was more effective than allowing participants to do it alone the entire time, as often they would not read the instructions. At the end of all experimental sessions we sent participants a survey which asked them a series of open-ended questions about their thoughts on the experiment and the different password entry methods. We use this qualitative data to add context to some of our results.

5 REPLICATION AND ANALYSIS OF PILOT ATTACK

5.1 Overview of Attack Replication Details

For our replication of the PILOT attack, we used existing datasets of digraph inter-key times (see [4, 5, 9]) rather than collecting our

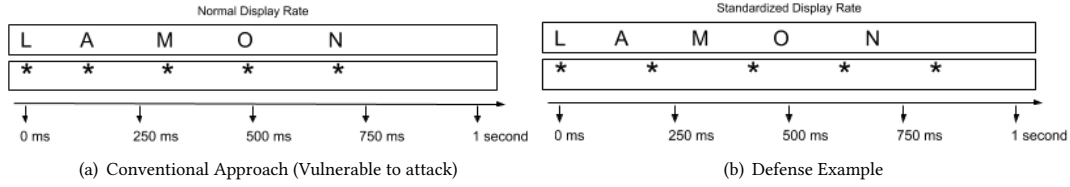


Figure 1: Figure on the left demonstrating how the attack works by allowing the attacker to find the time between obfuscated characters. Figure on the right demonstrating how a potential defense mechanism could protect against it by making the delay between characters indistinguishable.

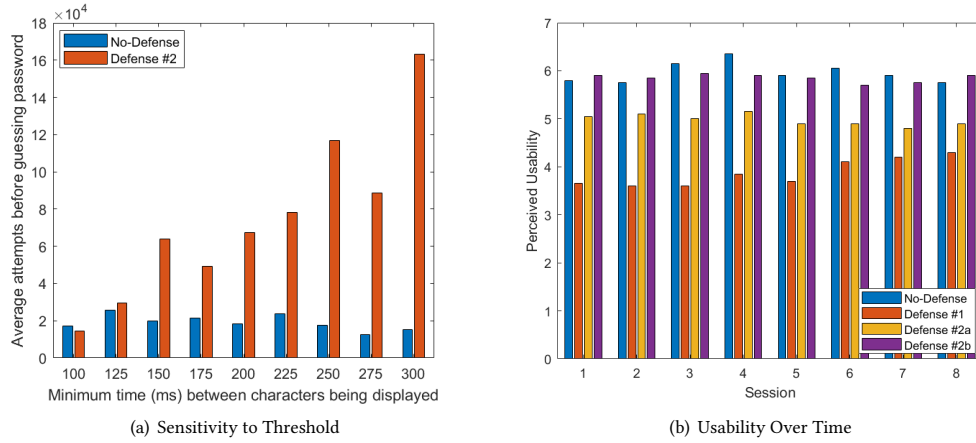


Figure 2: Left figure showing the number of guesses on average before a password (in this case “lamondre”) is correctly guessed by an attacker who is ranking passwords based on the DDT data for the password. Right figure showing the change in usability ratings for each password entry method broken down by the session in which the data was collected.

own. Like was done in [1], we also used the RockYou [6] dataset for ranking passwords in terms of how commonly they appear.

Our dataset preparation and model training followed the procedures in [1], and we replicated their method of scoring the ability of the model to guess the password. A variation of our attack evaluation from that in [1] was that we included a performance analysis of how the defense impacts the attack at different settings of the minimum threshold used by “Defense #2a” and “Defense #2b”.

Other key details for dataset handling and model training are that we used a 70/30 train/test split and the random forest was trained with default parameters in Scikit learn [8]. Digraphs with fewer than 100 samples were removed, those with more than 1000 were down-sampled. Results are based on 50 iterations of the experiment and an averaging of the results. Finally, we refer to what the authors call inter-keystroke timing as down-down time (DDT).

5.2 Attack Performance and Impact of Defense Threshold

Figure 2(a) shows the performance of the attack without the defense mechanism (blue bars), and for different threshold values of Defenses #2a or #2b (red bars). Observe that all blue bars reach just

about a value of 2×10^4 on the y-axis, meaning that it takes the attacker about 2×10^4 attempts to guess the password *if no defense is used*². With the defense in use, the number of required guessing attempts generally increases as the threshold value increases from 100ms towards 300ms. In general, when the threshold is high, this means that the proportion of digraphs whose inter-key times are manipulated/morphed increases, leaving the attacker with only a few digraphs that benefit from the attack. Figure 2(a) suggests that 250 ms or 300 ms would be the best threshold setting for our defense given these settings tremendously increase the attacker’s burden. After further analysis of which of these two values is more representative of users’ average typing speeds (results not shown due to space limitations), we chose the threshold 250ms for our usability experiments as it was better aligned with user’s natural typing rates.

²These numbers need to be interpreted in the context of an offline password guessing attack since online attacks would not permit such a number of guesses.

6 USABILITY OF DEFENSES AGAINST AUTOMATED VISUAL KEYSTROKE TIMING EXTRACTION

This section describes our results from the usability experiments. We estimate the usability that users associate with each of the more secure password entry methods (as compared to the conventional insecure approach) based on their qualitative and quantitative feedback, as well as password entry error rates. We did not use a specific method of evaluating our qualitative data for this paper and instead used it to give context to the quantitative data we collected from participants.

Password Entry Errors: Table 1 shows that users adapted poorly to losing the visual cues associated with the password entry process. The error rate for the method with no visual cues (“Defense #1”), had a password entry error rate nearly double that of the other defensive approaches and the unmodified “No-Defense” approach (9% vs 5-6%). One surprising aspect to these results was that there was not a significant change in the error rates for users when the password entry process included a delay when typing the password.

A possible explanation for this is in our qualitative data. Some of the responses suggest that many users might be using the visual cues of the asterisks to remember where they are at in a password being entered on a keyboard or help detect missed characters. For example, one participant stated that they “don’t like [Defense #1] since i’m quite forgetful and I prefer something visual.” We did have one participant that seemed enthusiastic about the new approach because it forced them to remain focused on the task, but this was the only such example of that sentiment. An opportunity for future research could involve exploring ways of making these defensive approaches more usable, similar to the work by [7] except focused on password delays (and/or lack of visuals) rather than partial obfuscation.

Usability Ratings: Table 1 also shows the mean and standard deviation of the Likert scale values from each participant over the course of the 8-day series of experimental sessions. Overall, the table confirmed our observation from the error rates that participants might react poorly to being forced to have no visual cues when typing in their password. However, although participants failed to enter the password correctly more often with the “Defense #2b” password entry method, they rated that approach higher than the “Defense #2a” approach (5.85 vs 4.99 out of 7). This is possibly due to some users having average DDT values that are much lower than the Defense #2a standardized rate, which would mean that the Defense #2b personalized rate would feel more natural to them (comparable no having no delay at all). By comparison the “Defense #1” approach scored a 3.87 and “No-Defense” scored a 5.96 out of 7.

Looking at Table 1, participants were generally in more agreement about the password entry approaches “No-Defense” and “Defense #2b”, with a standard deviation in the scores of 1.20 and 1.25 respectively. Defense #2a and Defense #1 had a standard deviation of 1.76 and 2.15 respectively. By examining the results more closely we found that some participants rated the “Defense #1” approach very highly, likely because they had prior experience with not having visuals when entering a password, but the average was dragged down by the majority of users.

Usability Over Time: Although only a single participant said that the “Defense #1” approach got easier over time, we did observe a steady improvement in the rating that “Defense #1” received over the course of the first 8 sessions (from about 3.6 to around 4.3 — see Fig 2(b)). The other password entry methods largely received the same rating over the course of the experiment.

To determine if our interpretation of the results was correct, we ran the Wilcoxon Signed Rank Test to determine whether the usability measurements for each password entry type come from the same population (for the first Vs last session). Our a-priori was set to 5% and we used standard settings (except for alternative set as “less”) when using the scipy python library’s implementation of the Wilcoxon test. Based on this configuration we found that the null hypothesis is not rejected for any of our password entry methods (i.e., no evidence that usability changed over time). Specifically the p-values were as follows. “No Defense” with 0.626, “Defense #1” with 0.081, “Defense #2a” with 0.744, and “Defense #2b” with 0.536. Given these results, which we discuss further in Section 7, we believe it would be interesting for future work to extend this study with a greater number of sessions and participants.

Other Qualitative Feedback: Based on the remainder of the qualitative data, we did find that most participants either couldn’t differentiate between “Defense #2a” and “Defense #2b” or they didn’t have a strong preference either way (e.g., “didn’t find them much different from each other”). One participant didn’t seem to like “Defense #2b” (“I did not like personalized because of the delay.”), but they were the only one. If such defenses were implemented at scale for millions of users, it might be worthwhile for a company to experiment with the different defensive approaches to customize the approach for a user based on what causes the least friction while also giving the most benefit. Such analysis might include error rates for the password entry process and data on the delay between characters (on the screen) showing that an attacker is unlikely to be able to extract much useful information from their target.

7 LIMITATIONS AND FUTURE WORK

We presented three password masking schemes designed to improve security against keystroke timing attacks, while minimizing changes to the widely used text-based password entry system. We evaluated the usability of the different schemes by controlling for the passwords and the platform (desktop). This approach may pose a limitation on how transferable our insights are in the case of mobile devices and other tactile platforms.

In addition, we collected usability data over only 8 sessions, which may not be enough to observe a convergence in our primary usability metrics, namely perceived usability (as measured by user sentiment/ratings) and practical usability (as measured by the change in errors made for each type).

Another limitation of our results is the demographics of our participants: all participants were either in college or at least had a college degree, which might correlate with greater technology proficiency. Future work could explore the role of demographics such as age, education, occupation, or primary language.

	No-Defense	Defense #1	Defense #2a	Defense #2b
Mean Error Rate	6.43	9.34	5.32	6.43
Mean Usability	5.96	3.87	4.99	5.85
Standard Deviation of Usability	1.20	2.15	1.76	1.25

Table 1: Table shows the error rate of password entry attempts, as well as the mean and standard deviation of the 7-point Likert scale responses we received for all password entry methods.

8 CONCLUSION

Our results show that although user sentiment was generally apposed to the “Defense #1” based approach of not having visual cues when entering a password, an organization could use different approaches for separate segments of their user population. Those who are more comfortable with technology and not having visual cues can opt out of them for the enhanced security, while everyone else could use threshold-based system of delays between characters being displayed on the screen to decrease the effectiveness of attacks that use key timing extraction techniques. Our experiments with the attack in Section 6 show that the threshold approach would still be a successful defense mechanism, while our usability analysis in Section 6 finds that most users are largely amenable to using a threshold to modify the display rate of characters on their screen (with “No-Defense” and “Defense #2b” having a 5.96 and 5.85 out of 7 rating respectively). This paper shows that there are acceptable options for the majority of users when it comes to enhancing security for the password entry process in areas where the visual cues for the user could be recorded or analyzed in aid of a future attack.

REFERENCES

- [1] Kiran Balagani, Matteo Cardaioli, Mauro Conti, Paolo Gasti, Martin Georgiev, Tristan Gurtler, Daniele Lain, Charissa Miller, Kendall Molas, Nikita Samarin, et al. 2019. PILOT: Password and PIN information leakage from obfuscated typing videos. *Journal of Computer Security* 27, 4 (2019), 405–425.
- [2] Alexander De Luca, Emanuel von Zezschwitz, and Heinrich Hußmann. 2009. Vibrapass: Secure Authentication Based on Shared Lies. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). Association for Computing Machinery, New York, NY, USA, 913–916. <https://doi.org/10.1145/1518701.1518840>
- [3] Alexander De Luca, Emanuel von Zezschwitz, Laurent Pichler, and Heinrich Hussmann. 2013. Using Fake Cursors to Secure On-Screen Password Entry. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 2399–2402. <https://doi.org/10.1145/2470654.2481331>
- [4] Romain Giot, Mohamad El-Abed, and Christophe Rosenberger. 2009. GREYC Keystroke: a Benchmark for Keystroke Dynamics Biometric Systems. In *IEEE International Conference on Biometrics: Theory, Applications and Systems (BTAS 2009)*. IEEE Computer Society, Washington, District of Columbia, USA, 1–6.
- [5] Romain Giot, Mohamad El-Abed, and Christophe Rosenberger. 2012. Web-based benchmark for keystroke dynamics biometric systems: A statistical analysis. In *2012 Eighth International Conference on Intelligent Information Hiding and Multimedia Signal Processing*. IEEE Computer Society, Washington, District of Columbia, USA, 11–15.
- [6] Kaggle. 2019. Common Password List. <https://www.kaggle.com/wjburns/common-password-list-rockyoutxt>.
- [7] Mohamed Khamis, Tobias Seitz, Leonhard Mertl, Alice Nguyen, Mario Schneller, and Zhe Li. 2019. Passquerade: Improving Error Correction of Text Passwords on Mobile Devices by Using Graphic Filters for Password Masking. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, Glasgow, Scotland Uk, 1–8.
- [8] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. 2011. Scikit-learn: Machine learning in Python. *the Journal of machine Learning research* 12 (2011), 2825–2830.
- [9] Joseph Roth, Xiaoming Liu, and Dimitris Metaxas. 2014. On continuous user authentication via typing behavior. *IEEE Transactions on Image Processing* 23, 10 (2014), 4611–4624.
- [10] Hirokazu Sasamoto, Nicolas Christin, and Eiji Hayashi. 2008. Undercover: Authentication Usable in Front of Prying Eyes. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) (CHI '08). Association for Computing Machinery, New York, NY, USA, 183–192. <https://doi.org/10.1145/1357054.1357085>
- [11] Abdul Serwadda and Vir V. Phoha. 2013. Examining a Large Keystroke Biometrics Dataset for Statistical-Attack Openings. *ACM Trans. Inf. Syst. Secur.* 16, 2, Article 8 (Sept. 2013), 30 pages. <https://doi.org/10.1145/2516960>
- [12] Dawn Xiaodong Song, David Wagner, and Xuqing Tian. 2001. Timing Analysis of Keystrokes and Timing Attacks on SSH. In *10th USENIX Security Symposium (USENIX Security 01)*. USENIX Association, Washington, D.C., 1–17. <https://www.usenix.org/conference/10th-usenix-security-symposium/timing-analysis-keystrokes-and-timing-attacks-ssh>
- [13] Susan Wiedenbeck, Jim Waters, Leonardo Sobrado, and Jean-Camille Birget. 2006. Design and Evaluation of a Shoulder-Surfing Resistant Graphical Password Scheme. In *Proceedings of the Working Conference on Advanced Visual Interfaces* (Venezia, Italy) (AVI '06). Association for Computing Machinery, New York, NY, USA, 177–184. <https://doi.org/10.1145/1133265.1133303>